Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

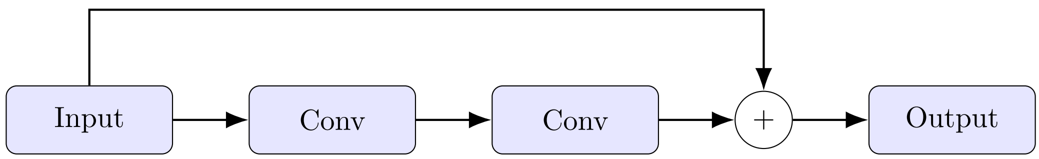

Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.
Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

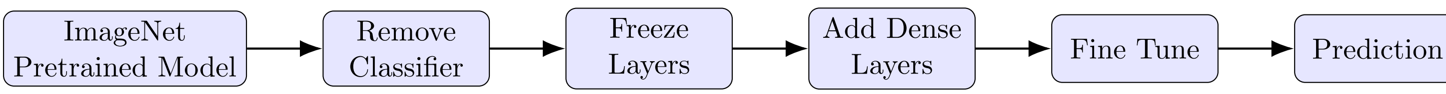

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy

- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Source Code

The source code and implementation are available in this GitHub repository:
GitHub Repository

18. Mandatory Plots

1. Sample CIFAR-10 Images

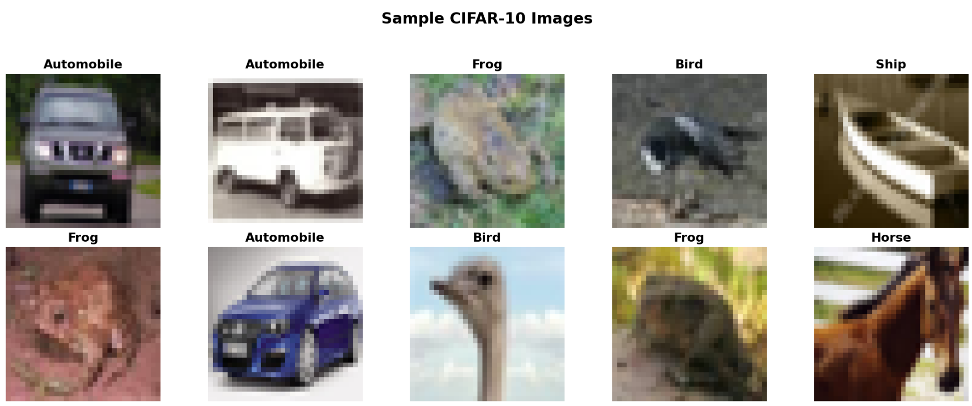

Figure 3: Sample images from the CIFAR-10 dataset.

Interpretation: The samples show different CIFAR-10 classes with clear variation in shape, colour, and background.

2. Training Accuracy

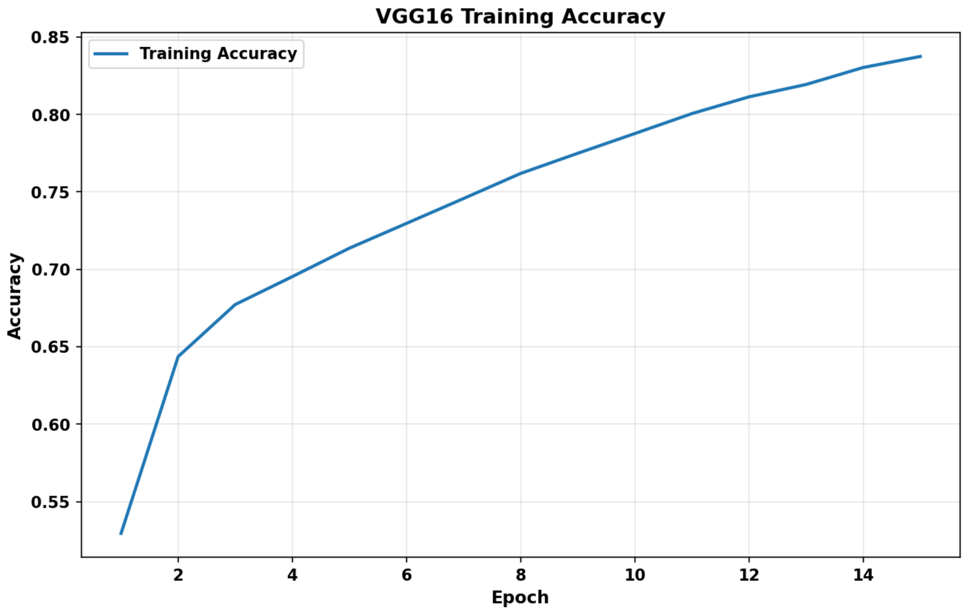

Figure 4: Training accuracy across the training epochs.

Interpretation: The training accuracy increases as the model learns more features from the training images. The curve shows how well the model is able to fit the training data over the epochs.

3. Validation Accuracy

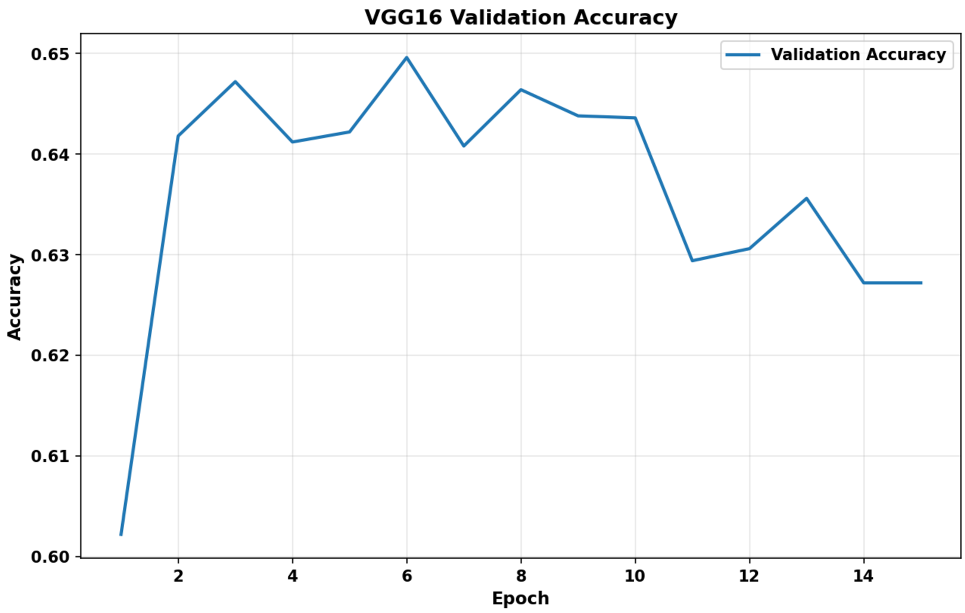

Figure 5: Validation accuracy across the training epochs.

Interpretation: The validation accuracy shows how well the model performs on images that were not used during training. The improvement in validation accuracy indicates that the pretrained model is able to transfer useful image features to the CIFAR-10 classification task.

4. Training Loss

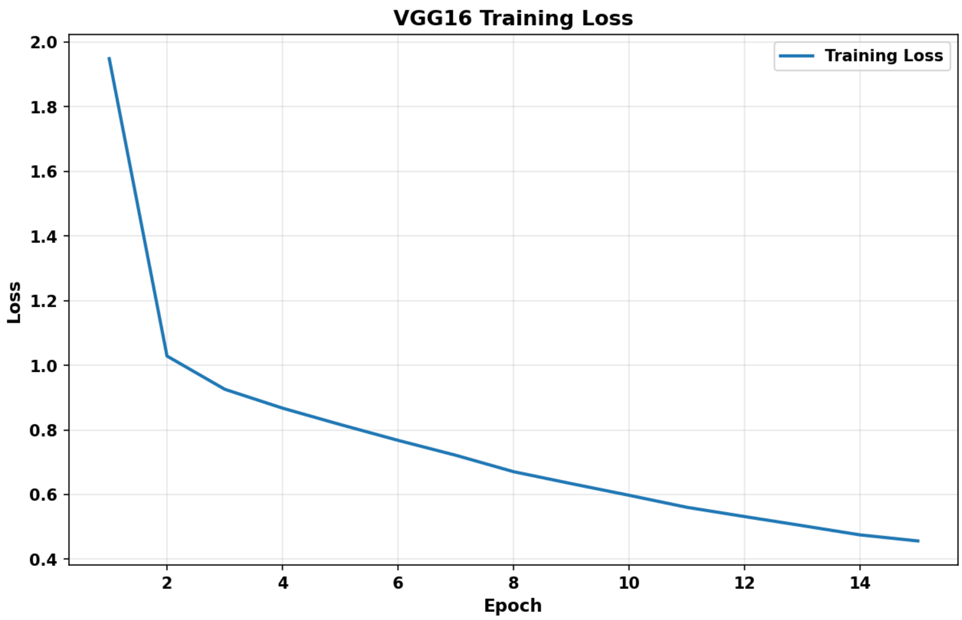

Figure 6: Training loss across the training epochs.

Interpretation: The training loss generally decreases as the model learns from the training data. A decreasing loss indicates that the model predictions are becoming closer to the actual class labels during training.

5. Validation Loss

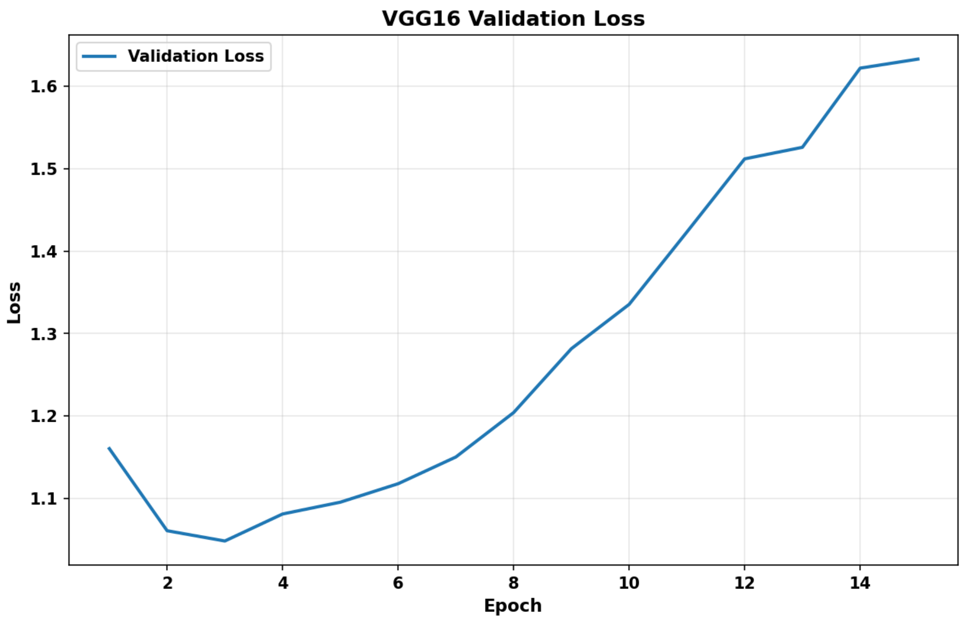

Figure 7: Validation loss across the training epochs.

Interpretation: The validation loss indicates how the model performs on unseen data. If the loss decreases along with the training loss, the model is learning useful features, while an increase after some epochs may indicate overfitting.

6. Confusion Matrix

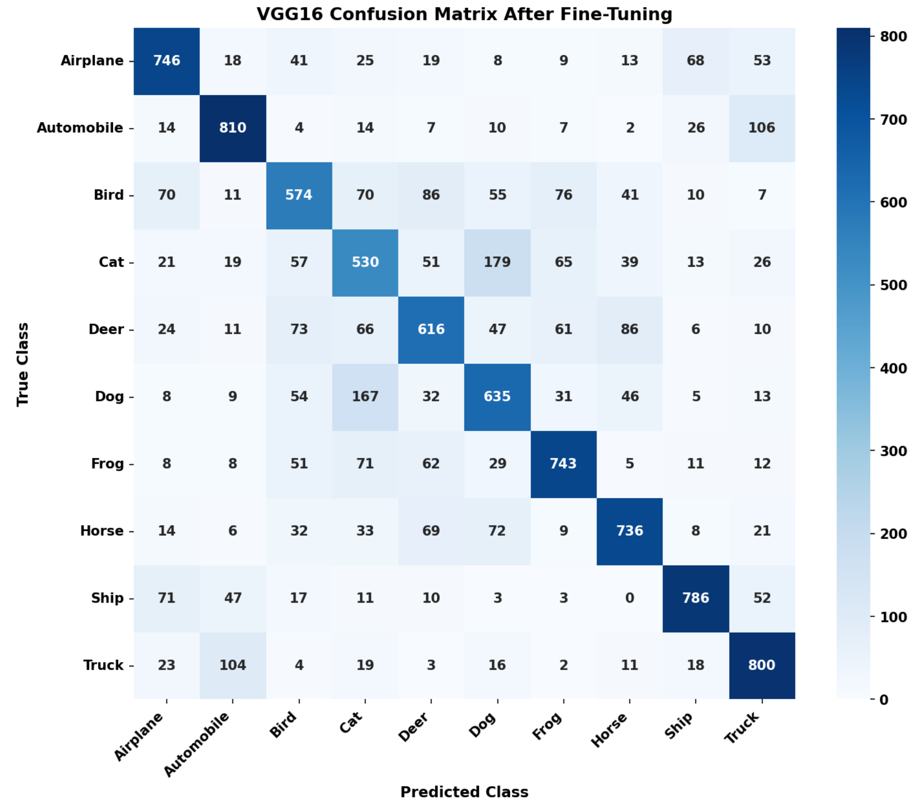

Figure 8: Confusion matrix of the trained transfer learning model on the CIFAR-10 test dataset.

Interpretation: The model performs well for classes such as automobile, airplane, frog, horse, ship, and truck, as shown by the strong diagonal values. However, some confusion can be seen between similar classes, especially cat and dog and between automobile and truck. This shows that the model finds visually similar classes harder to distinguish.

19. Results

19.1 Performance Metrics

Metric	Value
Best Training Accuracy	83.75%
Best Validation Accuracy Before Fine-Tuning	64.96%
Best Validation Accuracy After Fine-Tuning	69.82%
Testing Accuracy	69.76%
Precision (Macro)	69.75%
Recall (Macro)	69.76%
F1-score (Macro)	69.70%
Total Parameters	14,781,642
Initial Training Time (min)	3.18
Fine-Tuning Time (min)	2.49

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (min)
LeNet-5	83,126	51.86	0.54
AlexNet	36,925,322	10.00	2.72
VGG16	14,781,642	65.34	1.12
ResNet50	23,851,274	66.20	1.54

20. Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**
AlexNet was a major breakthrough because it showed that deep CNNs could perform very well on large-scale image classification tasks. It also made effective use of ReLU, Dropout, data augmentation, and GPU training, which helped make deeper CNNs practical to train.

2. **Why does VGG16 use only 3×3 convolution filters?**
VGG16 uses small 3×3 filters and stacks several convolution layers together. This allows the network to learn more complex features while keeping the number of parameters lower than using larger filters. The deeper layers can gradually learn features from simple edges to more complex patterns.

3. **Explain the advantages of the Inception module.**
The Inception module uses different filter sizes such as 1×1, 3×3, and 5×5 in parallel, along with pooling. This helps the network capture features of different sizes at the same time. The 1×1 convolutions also reduce the amount of computation and the number of parameters.

4. **What is the purpose of residual learning?**
Residual learning makes it easier to train very deep networks by using skip connections. Instead of learning the complete mapping $H(x)$ directly, the network learns the residual $F(x)$ and adds the original input to it. This also helps the gradient pass through the network more easily during training.

5. **Differentiate LeNet and ResNet.**
LeNet is a small and relatively simple CNN that was mainly designed for tasks such as handwritten digit recognition. ResNet is much deeper and was designed for more complex image classification tasks. The main difference is that ResNet uses residual or skip connections, which make training deep networks easier.

6. **What is Transfer Learning?**
Transfer learning means taking a model that has already learned useful features from a large dataset and adapting it to a new task. In this experiment, pretrained CNN models are used as a starting point for CIFAR-10 classification instead of training the entire network from scratch.

7. **Why is fine tuning required?**
Fine tuning allows some of the pretrained layers to update their weights according to the new dataset. After initially keeping the pretrained layers frozen, fine tuning helps the model learn features that are more specific to CIFAR-10 and can improve its performance.

8. **Explain the difference between dilated convolution and transpose convolution.**
Dilated convolution increases the receptive field without requiring a larger number of kernel parameters. It is useful when we want the network to capture information from a wider area of the image. Transpose convolution, on the other hand, is mainly used to increase the spatial size of feature maps through learnable upsampling.

9. **Why do pretrained models converge faster?**
   Pretrained models already have useful features such as edges, textures, and shapes learned from a large dataset. Therefore, they do not have to learn these basic features from the beginning. They can instead focus on adapting the existing features to the new task, which usually makes training faster.

10. **Compare the computational complexity of LeNet and ResNet.**
    LeNet is much smaller and has fewer layers and parameters, so it requires less computation and training time. ResNet is much deeper and has many more parameters, making it more computationally expensive. However, its residual connections allow it to train effectively and handle more complex image classification tasks.

21. Additional Exercises

1. **Implement transfer learning using VGG16.**
   VGG16 was used with pretrained ImageNet weights, and its convolutional layers were used to extract features. A new classifier was added for the CIFAR-10 classes. The model achieved a best validation accuracy of 65.34%.

2. **Repeat the experiment using ResNet50.**
   The same transfer learning approach was applied using ResNet50. It achieved a best validation accuracy of 66.20%, which was slightly better than the 65.34% obtained with VGG16.

3. **Compare Adam and SGD optimizers.**
   Adam performed better than SGD in the configurations tested. With a learning rate of 0.001 and a batch size of 32, Adam achieved 65.28% validation accuracy, while SGD achieved 62.62%.

4. **Compare training with frozen layers and fine tuning.**
   Before fine tuning, the model achieved a validation accuracy of 64.96% with the pretrained layers kept frozen. After fine tuning, the accuracy increased to 69.82%. This improvement shows that allowing the pretrained layers to adapt to CIFAR-10 helped the model perform better.

5. **Evaluate the model using Precision, Recall and F1-score.**
   The final model achieved a macro Precision of 69.75%, Recall of 69.76%, and F1-score of 69.70%. Since these values are quite close to each other, the model had fairly consistent performance across the different CIFAR-10 classes.

6. **Compare the performance of LeNet, AlexNet and ResNet.**
   ResNet50 achieved the highest validation accuracy of 66.20% due to its deeper architecture and residual connections, which help it learn complex features effectively. LeNet-5 achieved 51.86% because its smaller architecture has less capacity for the CIFAR-10 task. AlexNet achieved only 10.00% in our configuration, indicating that it did not adapt well to the dataset.

22. References

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6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
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